

Implementation of rPPG Signal Processing and Frequency Analysis for DocBot Heart Rate Extraction

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As part of the DocBot platform's health monitoring capabilities, the system utilizes Remote Photoplethysmography (rPPG) via RGB cameras to non-invasively detect periodic color fluctuations in the skin caused by blood volume changes. We have accumulated substantial strength in processing these raw optical signals into reliable, hospital-grade heart rate metrics. Look into the current implementation of rPPG routines. If the following has not been implemented, here are my comments on reserach directions. If some of the following are implemented, summarize what has been implemented; we may understand more and find new directions from there.

1: Filter Design and Signal Pre-Processing

- **Core Objective:** Design and apply **frequency-domain bandpass filters** to the extracted blood volume pulse signal. We must isolate frequency components that fall within a physiologically plausible heart rate range of **0.7 Hz to 3.5 Hz (approximately 42 to 210 beats per minute)**, while strictly attenuating noise and respiratory artifacts.
- **Methodology:** Evaluate and compare **Finite Impulse Response (FIR)** filter designs (such as windowed sinc or Parks-McClellan optimal equiripple filters) against **Infinite Impulse Response (IIR)** filter designs (such as Butterworth, Chebyshev, or elliptic filters). The goal is to configure the cutoff frequencies and filter orders to optimize the sharpness of the transition bands.
- **Lab Reference:** Use the **Filter Design Toolbox in MATLAB** to generate these filters. See an example in **Figure 4 in our lab's European Journal of Control paper: [1]**. Xu Chen and Masayoshi Tomizuka, 'Selective model inversion and adaptive disturbance observer for time-varying vibration rejection on an active-suspension benchmark'. European Journal of Control, 19(4):300 - 312, 2013. <http://dx.doi.org/10.1016/j.ejcon.2013.04.002>.

2: Frequency Domain Analysis

- **Core Objective:** Build frequency domain analyzers to understand the exact heart rate from your filtered signals.
- **Methodology:** Transform the time-domain rPPG signal into a frequency-domain representation (power spectrum) using techniques like **Fast Fourier Transform (FFT)**, **power spectral density estimation via Welch's method**, or **Short-Time Fourier Transform (STFT)**.
- **Implementation:** Compute this analysis over a **sliding temporal window of different lengths**. The macs-matlab-toolbox has provided the function specCal and specCale to calculate the spectrum.

3: Rapid Peak Detection

- **Core Objective:** Extract the exact heart rate from your filtered signals rapidly, ideally using much less frames of data compared to the state of the art.
- **Methodology:** Frequency identification algorithms such as MUSIC and ESPRIT are excellent at the job. See attached my summary and example illustration of the MUSIC algorithm. I have also published a sample code and example in this github repository.
- **Implementation:** Compute this analysis over a **sliding temporal window of different lengths**. Identify the minimum needed data to get an accurate frequency value and benchmark against the state of the art. Your peak detector must identify the frequency bin with the maximum power spectral density within your filtered 0.7–3.5 Hz range. Convert this dominant peak frequency into beats per minute (BPM) by multiplying the Hz value by 60.

4: Confidence Scoring and Sensor Orchestration

- **Core Objective:** Because DocBot operates in dynamic environments, the AI must be able to self-evaluate the reliability of the heart rate data it collects.
- **Methodology:** Design a **confidence score** for the extracted heart rate measurement. Calculate the score based on a well thought out set of parameters, e.g.:
 1. The **Signal-to-Noise Ratio (SNR)** of the dominant cardiac frequency peak relative to the surrounding frequency bins.
 2. The **temporal stability** of the heart rate estimate across successive sliding analysis windows.
 3. The **available skin pixel count** used for spatial averaging from the RGB camera feed.
- **System Integration:** Integrate this confidence metric into the DocBot's overarching **orchestration module**. The module will use your score to assess signal reliability and determine if the robot needs to trigger **sensor repositioning** to the feedback robot control part, e.g., improve the camera's view of the patient and boost signal quality.